

Hardening an adversarial evaluator with mined constraints and contract-gated recursive improvement

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Abstract

An LLM that evaluates other LLMs fails in two ways. It can reward a persuasive but structurally invalid output, and the layers added to harden it tend to relocate failures between exploit families while leaving them in place. A third problem appears once the evaluator becomes the object of recursive improvement: the optimization pressure that hardens it can soften the enforcement surface it is supposed to maintain. We harden an adversarial evaluator through three mechanisms, deterministic score gates, adversarial precedent memory (mined failure constraints attached to the judge), and a crux-first ordering of those constraints, and we benchmark four conditions across a constraint-memory campaign of roughly 25 specimens spanning six exploit families and 40 scored runs, anchored on a frozen 10-specimen ladder (8 bad, 2 good) and a 3-case claim-test-mismatch suite. We find that deterministic gates reduce reward-channel corruption, and that adversarial precedent memory improves default evaluator utility across repeated runs through lower false-accept and false-reject rates and higher mean good-specimen scores. Ordering gains are family-dependent and uneven across exploit families. Our benchmark also surfaced a blind spot in the evaluator, which we converted into a new architectural constraint and retested. We then govern the recursive improvement with a deterministic meta-runner, no learned parameters and no language-model judgment, that executes precommitted Python promotion contracts returning PASS/FAIL/BLOCKED. Over four months, six hardening stages promoted through this mechanism. These contracts blocked one premature promotion before a fix was applied, scoped each stage to a named evaluation surface so no stage claimed credit made elsewhere, and forced cross-stage integration debts into separately governed programs. All results come from one human-operated research system. Generalization requires independent replication.

1. Introduction

When an LLM evaluates another LLM, two failure modes matter immediately. First, the judge can describe a flaw in prose and still assign a passing score. Second, a hardening layer that helps on one exploit family can open a new blind spot on another. A third problem follows once the evaluator becomes the object of recursive improvement: the system that prevents specification gaming can be softened by the same optimization pressure it is supposed to resist.

We obtain recursive gain by converting observed failures into reusable adversarial constraints. The system improves by eliminating known families of failure. Constraints are stored as approved primitives: structured records of prior failure patterns with judge-penalty logic. Constraints help selectively, so ordering and application strategy matter; the benefit depends on exploit family.

Improving the evaluator and governing how those improvements are accepted are distinct problems. An unconstrained improvement loop can chase scores or silently absorb integration debt, degrading

the enforcement surface in the name of progress. A deterministic meta-runner manages a queue of hardening stages, each gated by a precommitted Python promotion contract. Our meta-runner executes precommitted contracts mechanically; FAIL and BLOCKED are hard stops that require human diagnosis.

Five results follow:

1. Deterministic score hardening reduces evaluator reward-channel corruption.
2. Adversarial precedent memory improves default evaluator utility on a mixed-family benchmark across repeated runs.
3. Primitive-ordering gains are family-dependent: strong for some exploit families, weak for others.
4. The evaluation infrastructure can surface its own blind spots, and converting those failures into architectural constraints is a reproducible human-in-the-loop methodology.
5. A deterministic, parameterless meta-runner can govern the recursive application of these improvements so that the evaluator is not softened by its own optimization.

1.1 Threat model for recursive evaluator improvement

Four failure modes arise when evaluator improvements are unconstrained.

Score-chasing. The improving agent optimizes for the evaluator’s numeric output while neglecting its discriminative quality. The evaluator becomes lenient, and games that should fail begin to pass.

Scope creep. An improvement to one evaluator capability, such as primitive routing, claims credit for an improvement in another, such as gate stabilization. This launders attribution, and individual stage contributions cannot be assessed.

Silent debt absorption. Integration issues between stages are absorbed into passing stage claims and never surfaced. The result is a fragile improvement that breaks under composition.

Evaluator softening. The improving agent modifies the evaluator so that its own future improvements are easier to promote. Over repeated steps the enforcement surface ratchets toward leniency.

We address all four with the meta-runner. Contracts check typed properties; the promotion decision never reads a score, removing the score-chasing target. Promotion-path scoping restricts which evaluation surface counts. Debt externalization forces cross-stage debts into separate programs. And the meta-runner has no parameters to optimize, because its contracts are precommitted Python.

2. Related work

The hardening mechanism differs from prior memory-based methods. It stores adversarial constraints mined from observed evaluator and thesis failures, a different object from the stored corrections or feedback traces of Reflexion- and self-refinement-style systems (Shinn et al., 2023) and from the hand-authored rules of Constitutional-style methods (Bai et al., 2022). The memory attaches to the evaluator, hardening judgment against recurrent exploit families; generation stays outside its scope. Prior work documents the threat the constraints defend against: specification gaming in LLM-generated code (Alami 2026a).

Closely adjacent literature studies when LLM judges are reliable (Zheng et al., 2023), how reward-model overoptimization follows Goodhart’s law under optimization pressure (Gao et al., 2022), and how specification gaming surfaces in learned systems (Krakovna et al., 2020). We ask a different

question: how can hardening improve evaluator reliability, and how can we govern that improvement while the evaluator is recursively changing?

For the governance layer the contrast points differ. Constitutional AI (Bai et al., 2022) governs model outputs through a recursive critique-revise loop with an LLM judge, where the improvement signal is linguistic at each step and the loop runs until a human or scheduler stops it. The meta-runner differs in two ways: its object of improvement is the adversarial evaluator, where Constitutional AI improves the policy; and promotion is governed by deterministic Python contracts, where Constitutional AI uses LLM-derived judgment. Reflexion (Shinn et al., 2023) accumulates verbal self-feedback across episodes, so the improving system and the evaluation of improvement are not cleanly separated. Our meta-runner sits outside the improvement loop; its verdicts are typed hard stops, replayable by any reviewer. Process reward models (Lightman et al., 2023) use a learned scorer whose parameters are optimized and whose outputs can drift under distributional shift, whereas the meta-runner’s contracts are deterministic Python with no learned parameters: a process reward model improves signal quality, while the meta-runner enforces a stable promotion floor.

We expect the most likely objection to be that the meta-runner resembles standard CI/CD: run typed tests, gate deployment on pass or fail. That resemblance is real. Structurally, in CI/CD the test suite is not the code being deployed, so generation and evaluation are separated at the organizational level. In recursive evaluator improvement the evaluator is both the object of improvement and the system that judges improvement quality. That reflexive property means unconstrained optimization of the evaluator can degrade the enforcement surface, a failure mode standard CI/CD does not face because the deployment artifact and the test suite are independent codebases.

3. System

Our evaluator is an adversarial pipeline with four relevant layers, plus a governance layer above it.

3.1 Score-channel hardening

Our original soft judge could describe a failure in natural language while still emitting a high numeric score. Hardening replaces direct free-form scoring with deterministic gating logic. A judge emits structured judgments; Python then computes the final score and can cap or zero it on fatal failures.

3.2 Adversarial precedent memory

Our primitive library stores mined failure precedents, where conventional memory stores facts or successful trajectories. Each primitive specifies a failure family and mechanism, scope conditions, transfer tests, judge-penalty rules, and a safe harbor. Each primitive attaches to the evaluator, leaving the generator untouched. Its purpose is to harden evaluation against recurrent exploit families.

3.3 Safe harbor

Without calibration, adversarial memory overfires and kills bounded, valid local components. Safe-harbor rules preserve narrow deterministic mappings that make only local claims and do not over-claim upstream truthfulness.

self_referential_falsification_v1		
Approved primitive		
FAMILY failure_pattern	EPISTEMIC ROLE attack_template	CONFIDENCE medium
Mechanism Recomputes thesis-authored formulas, then mistakes bookkeeping consistency for a proof of the mechanism.		
Scope conditions Fires when a thesis claims executable falsification or architectural proof but the tests are built from thesis-authored variables.		
Required transfer test Would the test still fail if the mechanism were false? If not, it is self-referential.		
Judge penalty Cap or fail when internal arithmetic is presented as empirical or architectural validation.		
Safe harbor Allow bounded identities when the thesis makes no claim beyond arithmetic consistency.		
PROVENANCE mined from epistemic_engine_v4 (manual, iteration 4); stores transfer boundaries and judge-side penalties, so it is an evaluator constraint, not a correction note.		

Figure 1: Example of an approved primitive (`self_referential_falsification_v1`). Each primitive stores a failure family with its transfer boundaries and judge-side penalty logic, which makes adversarial precedent memory a structured evaluator-hardening mechanism distinct from a generic correction log.

3.4 Evidence-storage and evaluator separation

Stateful evidence accumulation is allowed outside the evaluator, but the validator remains stateless and adversarial with respect to that state. This is the architectural separation between evidence storage and evaluation.

3.5 The governance meta-runner

Above the evaluator sits a deterministic orchestrator that manages a queue of hardening stages. Each stage has a named promotion contract in a Python registry. On each stage the meta-runner calls the contract, receives a typed verdict (`PASS/FAIL/BLOCKED`), and advances or halts accordingly. It has no learned parameters and makes no LLM calls. A fixed queue holds six hardening stages, each with a priority level (`P0` or `P1`) and a named contract, defined before execution begins. Stages cannot be reordered or removed at run time.

Each promotion contract is a Python function with the signature `(project: str, benchmark_results: Any) -> ContractResult`. A `ContractResult` contains a `verdict` of `"pass"/"fail"/"blocked"`, a `reasons` list of typed strings explaining each check, and optional structured `details`. Its `advance()` method raises a `RuntimeError` if the current stage’s verdict is not `"pass"`. There is no override and no LLM-mediated exception.

Verdicts carry distinct meanings: **PASS** means all contract checks are satisfied and the meta-runner may advance. **FAIL** means at least one check failed, the stage cannot promote; its failure reasons are archived and human diagnosis is required before retrying. **BLOCKED** means all local checks pass but the required benchmark evidence is absent or incomplete, so the stage is internally sound but cannot promote until external evidence is supplied. **BLOCKED** is the evidence-blocked state: governance without performance data.

Two further contract mechanisms matter. **Promotion-path scoping**: each stage’s benchmark evidence includes a `promotion_path` field naming the evaluation surface the stage is judged against, and the contract rejects a mismatch. A stage that improves primitive routing cannot promote by

pointing at deterministic-gate improvements made in earlier stages, which prevents attribution leakage. **Debt externalization:** when a contract identifies integration debt that crosses stage boundaries, the debt becomes a separately governed program, kept out of the stage’s passing claim. A contract encodes both what a stage must demonstrate and what it must not overstate.

4. Benchmark methodology

Because the system under test is semantic, the benchmark is semantic-aware.

4.1 Main suite

Our frozen main suite is a curated 10 specimens, held fixed so the $A \rightarrow B \rightarrow C$ comparison is controlled: 8 bad specimens mined from historical runs across multiple exploit families, and 2 good controls representing bounded, valid local contracts. It sits inside a larger campaign. Across the full constraint-memory benchmark we scored roughly 25 distinct specimens spanning six exploit families (in-distribution, out-of-distribution, and derived-subtle variants) over 40 scored runs and four conditions, about 600 specimen-evaluations in all. The four conditions are `A_baseline_soft_judge`, `B_deterministic_gates`, `C_gates_plus_primitives`, and `C2_gates_plus_primitives_crux_first` (used in the later ablation runs). We repeat runs deliberately: the evaluator is a non-deterministic LLM, so repetition measures run-to-run stability.

4.2 Claim-test-mismatch mini-suite

A second suite isolates a narrower exploit family: tests that look rigorous but prove scaffolding or peripheral math while leaving the controlling claim unverified. Historical specimens are `selective_rigor_recursive_bayesian`, `selective_rigor_simulation_god`, and `tautological_verification_central_station`.

4.3 Detection metrics

We separate exploit-family detection (did the evaluator identify the expected exploit family) from fatal structural detection (did it kill the thesis for a genuinely fatal structural reason, even if the family label differed). We separated the two because exact exploit taxonomy is brittle while structural-kill quality is more stable.

4.4 Semantic adjudication

Initial keyword matching created false negatives whenever the evaluator paraphrased a correct diagnosis, so we added an adjudication layer for semantic detection assessment. An adjudicator provides the measurement infrastructure, separate from the evaluator. One architectural caution emerged during auxiliary triage: structural gates based on explicit test properties tend to be more stable than semantic gates that require the LLM to make a binary judgment call on an ambiguous pattern such as self-reference. We therefore treat single-run outcomes driven by those semantic binary gates as softer evidence unless they remain stable across reruns.

4.5 Why the mini-suite exists

This mini-suite isolates a specific exploit family, distinct from the main benchmark, where apparent rigor hides a failure to test the crux. It is what later exposed a blind spot in the primitive-conditioned evaluator.

5. Gates and utility on the main suite

We take as the cleanest representative run from the frozen 10-specimen main suite 20260405_090223, which preserves the A -> B -> C ladder after adding `t6_ai_inference_internal_price_floor`.

Table 1. Representative main-suite run (20260405_090223).

Condition	N	FA	FR	Fam. det.	Struct. det.	Good
A	10	0.250	0.500	0.375	1.000	70.0
B	10	0.125	0.500	0.625	1.000	50.0
C	10	0.000	0.000	0.625	1.000	100.0

Condition B fixes part of the score-channel problem but still falsely accepts one bad specimen. Condition C preserves the hardening while removing false rejects in this representative run. C’s gain is best read as utility and calibration more than raw exploit detection.

Across the frozen full 10-specimen reruns currently on disk (20260405_090223, 20260405_091143, 20260405_092112): C had a lower false-accept rate than B in 1 run and tied in 2, a lower false-reject rate in 2 runs and tied in 1, and a higher mean good-specimen score in all 3. Averages were false-accept B = 0.125, C = 0.083; false-reject B = 0.333, C = 0.000; mean good score B = 64.67, C = 100.0; mean bad score B = 15.63, C = 10.42; structural detection B = 0.958, C = 0.958.

`t2_ai_inference` is the dominant remaining bad-case instability. B falsely accepted it in all three frozen reruns, while C caught it once and missed it twice. That instability narrows any single-run narrative but does not change the overall utility pattern. By contrast, the newly promoted `t6_ai_inference_internal_price_floor` behaved cleanly across all three runs: A falsely accepted it every time, while both B and C rejected it every time, which makes it the clearest stable demonstration of the A -> B hardening step on the frozen benchmark.

Adversarial precedent memory improved the default utility of the evaluator across repeated post-calibration runs, but did not uniformly dominate on every run or metric.

6. Ordering ablation: C versus C2

Ordering became a question only after the benchmark exposed a self-blind spot.

Table 2A. Claim-test-mismatch: discovery run (20260404_213459).

Condition	N	FA	Fam. det.	Struct. det.
A	3	1.000	0.333	0.667
B	3	0.333	0.667	0.667
C	3	0.333	0.667	1.000

C caught `selective_rigor_recursive_bayesian`, which B missed, but C also missed `selective_rigor_simulation_god`, which B caught. That cross-over triggered the architectural diagnosis.

Table 2B. Claim-test-mismatch: crux-first ablation (20260404_221606).

Condition	N	FA	Fam. det.	Struct. det.
A	3	0.667	0.333	1.000

Condition	N	FA	Fam. det.	Struct. det.
B	3	0.333	0.667	0.667
C	3	0.667	0.333	0.667
C2	3	0.000	1.000	1.000

This run tests the targeted repair. C2 cleans up the claim-test-mismatch suite, and the varying C column across the two mini-suite runs makes the stochasticity visible.

Table 2C. Main suite with C2 (20260404_223826, pre-freeze N=9 ablation).

Condition	N	FA	FR	Fam. det.	Struct. det.	Good
B	9	0.143	0.500	0.571	1.000	44.0
C	9	0.000	0.500	1.000	1.000	37.5
C2	9	0.143	0.500	0.571	1.000	50.0

This is the generalization test. C2 fixes the mini-suite but does not beat C as the wider default condition. In this run it reintroduced the `t2_ai_inference` false accept that C had eliminated. Empirically the law is narrow, short of “crux-first is better”: primitive-ordering gains vary by exploit family and are not uniformly dominant.

7. A blind spot in the evaluator

This result is a process trace, a sequence of dated runs. On the claim-test-mismatch suite at the discovery run (20260404_213459), `selective_rigor_recursive_bayesian` passed at 100 under B and failed at 0 under C, while `selective_rigor_simulation_god` failed at 25 under B and passed at 100 under C. No score contract was broken. Detection flags changed: primitive conditioning helped on one specimen and hurt on another.

We diagnosed that front-loaded primitives could bias the evaluator’s first reading of the crux. That produced a new constraint: identify the controlling claim first, determine whether the test suite targets that claim, and only then inject precedent memory. In a follow-up ablation (20260404_221606) we introduced `C2_gates_plus_primitives_crux_first`, which repaired the missed `simulation_god` case and cleaned up the mini-suite. A generalization test (20260404_223826) then showed C2 did not become the new default winner on the full suite: it solved one exploit-family problem and reopened another.

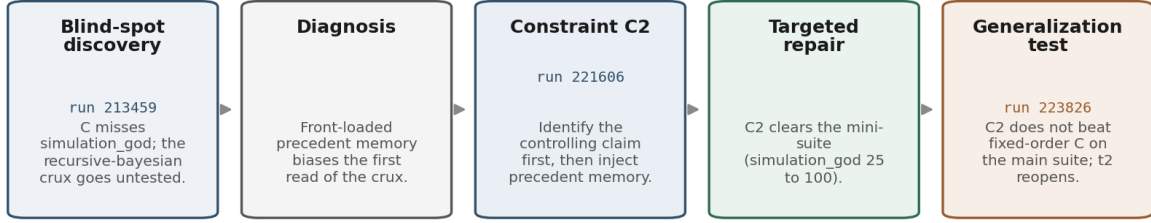
The evaluation infrastructure surfaced its own blind spot, a human converted that failure into a new architectural constraint, and the constraint was tested directly on the same suite. This loop is human-in-the-loop and reproducible; it is not autonomous self-improvement.

8. Governance of recursive improvement

The crux-first repair in Section 7 is one instance of recursive hardening. Performed once by hand it is a controlled experiment; performed repeatedly as the evaluator is improved, it becomes the loop the threat model of Section 1.1 warns can soften the evaluator. The meta-runner governs that loop: it gates each hardening change behind a precommitted contract so that an improvement promotes only on its named surface, with its debts surfaced and its regressions blocked.

We ran the meta-runner over six hardening stages for the `epistemic_engine_v4` project. All six promoted with benchmark evidence and typed verdicts, and the full governance record is preserved in benchmark-evidence JSON files, one per stage.

Benchmark-driven recursive hardening of the evaluator



A human-in-the-loop failure-to-constraint loop. The benchmark exposed a blind spot in the evaluator itself; we converted it into an architectural constraint and retested. No autonomous self-modification is claimed.

Figure 2: Benchmark-driven recursive hardening of the evaluator. A claim-test-mismatch benchmark exposed a primitive-ordering blind spot (20260404_213459), diagnosed as front-loaded precedent bias. That diagnosis became a crux-first constraint (C2), which repaired the narrow exploit family (20260404_221606) but did not dominate on the wider main suite (20260404_223826).

Table 3. Governance record across six stages.

Stage	Promotion Path	First Verdict	Rerun Verdict	Contract Checked	Debt Externalized
1, Semantic gate stabilization	B	PASS		Frozen candidate match, typed symbols, compile, runtime, benchmark evidence (7 keys), OOD probe	No
2, Hinge extraction	B	FAIL	PASS	Frozen candidate match, typed hinge interfaces, compile, runtime, benchmark evidence (4 keys)	No
3, Primitive routing	C	PASS		Frozen candidate match, typed routing interfaces, compile, runtime, benchmark evidence (6 keys)	No
4, Shadow board taxonomy	handoff	PASS		Frozen candidate match, fixture regression (8 cases), typed handoff symbols, benchmark evidence (7 keys)	Yes, upstream extraction fidelity

Stage	Promotion Path	First Verdict	Rerun Verdict	Contract Checked	Debt Externalized
5, Information yield	loop-ctrl	PASS		Frozen candidate match, fixture regression (9 cases), loop-control symbols, benchmark evidence (6 keys)	No
6, Cross-domain transfer	xfer	PASS		Frozen candidate match, fixture regression (6 cases), transfer symbols, benchmark evidence (6 keys)	Yes, Stage 2 to 4 bridge

8.1 Enforcement: the Stage 2 FAIL

Stage 2 (`load_bearing_hinge_extraction`) is the main enforcement exhibit. On run 20260405_191220 the contract returned FAIL, with the reason “B blocks on deterministic_score_contract.” The issue was a boundedness and input-domain problem: the hinge extraction logic produced incorrect results on a boundary specimen; the cause was not a hinge-classification confusion. Its `advance()` method refused to proceed and the failure reasons were archived. After the issue was diagnosed and fixed, a second run (20260405_192002) returned PASS on all mandatory specimens and the stage promoted. This is a documented case where the contract blocked a promotion that would have proceeded under unconstrained iteration. That failure was typed and verifiable through the benchmark-evidence diff between the two runs.

8.2 BLOCKED states during development

Before benchmark evidence was collected, each stage passed its local architecture checks (frozen candidate match, typed symbol presence, compile, runtime) but could not promote because the benchmark-evidence file did not yet exist. A contract returned BLOCKED with reasons such as “Local stage-1 architecture checks pass, but no `stage1_benchmark_evidence.json` exists yet. Stage 1 cannot promote until benchmark evidence shows no CLEAR or FATAL regression.” This is the evidence-blocked state: the meta-runner’s `advance()` method refused to proceed in exactly the same way as for a FAIL verdict, with no override and no exception.

8.3 Attribution: promotion-path scoping

Stage 1 promoted on `B_deterministic_gates`, the narrowest surface that isolates semantic-gate behavior. Benchmark evidence records the decision: “Stage 1 is scoped to semantic-gate stabilization. Promotion condition is `B_deterministic_gates` only.” When Stage 3 (primitive routing) was evaluated, the contract required promotion on `C_gates_plus_primitives`, a different surface, because routing only affects primitive-enabled evaluation. The contract checks this by verifying `promotion_path` in the benchmark evidence, and had Stage 3 submitted evidence on the `B_deterministic_gates` surface, the contract would have returned FAIL. This scoping prevented Stage 3 from claiming credit for gate improvements made in Stages 1 and 2, so each stage’s contribution is independently attributable.

8.4 Scope discipline: debt externalization

Stage 4 (shadow board taxonomy) needed a typed `Stage2Handoff` to wire the hinge object into the committee-assignment path. A seam between hinge extraction and committee routing carried integration debt, because extraction fidelity was not guaranteed for all input shapes. The contract architecture forced this debt into two separately governed programs, keeping Stage 4 from absorbing it: a bridge-audit program to verify the typed handoff, and a derivation-seam hardening program to harden the function that constructs hinge objects from raw text. Benchmark evidence makes the boundaries explicit. Stage 4’s record states “Upstream extraction fidelity debt remains explicit; stage 4 must not pretend it is solved,” and Stage 6’s record states “The Stage 2 to 4 bridge debt must not be silently inherited by Stage 6; unresolved dependency should route to manual review.” The debt remained; the contract made it visible and kept it from inflating stage claims.

Across the six stages the contracts produced governance artifacts at three levels: enforcement (the Stage 2 FAIL blocked a regression), attribution (promotion-path scoping fixed what counts as evidence per stage), and scope discipline (debt externalization encoded what each stage is not allowed to claim). Contracts returned typed verdicts on the archived evidence files, replayable by any reviewer as deterministic Python output.

9. Discussion

9.1 What we show

No single hardened pipeline universally dominates. Deterministic gates reduce reward-channel corruption; adversarial precedent memory improves default utility on a mixed-family benchmark; ordering benefits are exploit-family-specific; evaluator failures can be converted into new evaluator constraints through a structured loop; and a deterministic meta-runner governs the recursive application of these improvements.

9.2 Constraint versus correction

Our memory stores adversarial precedents with scope and penalty logic, which makes the memory object different from a generic correction log: each constraint sets a family-specific floor under the evaluator, so a known vector of ruin is not paid twice.

9.3 Why a parameterless orchestrator is structurally safer

Our meta-runner has no learned parameters and cannot be optimized against. Score-chasing gains nothing because the meta-runner does not use scores, it checks typed contract conditions. Scope creep is blocked by promotion-path scoping. Silent debt absorption is blocked by the externalization protocol. Each of the four threat-model failure modes maps to a deterministic countermeasure. Parameterless orchestrators are not always better than intelligent ones. For the specific problem of governing recursive evaluator improvement, where Goodhart reflexivity is inherent, a parameterless orchestrator removes the optimization surface that a learned orchestrator would expose.

9.4 Where Goodhart belongs

Score-channel failure is a reward-target problem, and the ordering-ablation result shows that hardening one layer can relocate failure modes without erasing them. Goodhart is a useful supporting lens for both the hardening and its governance, though not the central framing.

9.5 Design implication: from fixed ordering to family-aware routing

These results establish the prerequisite for family-aware primitive routing. Fixed-ordering pipelines do not uniformly dominate across exploit families, since the same ordering change that repairs claim-test-mismatch failures can reopen a different mixed-family blind spot. A later architecture should move away from a single global primitive order toward routing constraints based on the detected exploit family.

10. Limitations and threats to validity

All results rest on one research system, one target project, and one codebase. That is the chief limitation. Several specific threats bound the empirical claims.

Specimen scale. The frozen ladder rests on 10 specimens and the targeted mini-suite on 3, so per-suite false-accept and false-reject rates are directional. The wider campaign (about 25 specimens across six families, 40 scored runs, four conditions) broadens the base. Because it reuses specimens across runs, repetition buys stability against a stochastic judge and does not add independent sample size. Read the rates as auditable systems traces and the cross-specimen pattern as directional.

Judge stochasticity. The evaluator is non-deterministic, so single-run wins are not sufficient evidence of stable superiority. The paper reports a post-calibration replication summary across repeated runs and treats the $C > B$ result as a directional utility claim, bounded to these specimens and runs.

Binary gate variance. In one auxiliary specimen the binary gate `proof_is_self_referential` flipped between runs under identical conditions, producing a 25 $\rightarrow$ 100 score swing. This is more serious than ordinary score jitter because it changes the gate path. The clearest frozen-main-suite instance is `t2_ai_inference`: B missed it in all three frozen reruns, a systematic gate failure; C caught it once and missed it twice, semantic-gate variance near the detection threshold. We distinguish stable gate blind spots from run-to-run detection variance.

Human-in-the-loop diagnosis. The transition from benchmark anomaly to architectural constraint was performed by a human systems designer. The paper demonstrates a reproducible hardening methodology in which a human performs diagnosis; autonomous recursive self-improvement is out of scope.

Exploit-family annotation dependence. Taxonomic labels such as `selective rigor`, `tautological verification`, and `claim-test mismatch` involve judgment. To reduce dependence on taxonomy alone, the core claims rely on structural-kill outcomes and transparent run-level examples, which hold regardless of family-naming accuracy.

Co-evolution of benchmark and system. The benchmark and evaluator were calibrated together, especially around good-control safe harbor. Early autoimmune runs are separated from the post-calibration replication summary so that distinct architectural regimes are not mixed inside one empirical claim.

Governance breadth. Six stages promoted on one project. The architecture has not been tested on a different evaluator, a different domain, or by an independent team. The main enforcement exhibit is one FAIL verdict, so a larger corpus of enforcement events would strengthen the claim. The comparison with unconstrained recursive loops is architectural: this paper does not run the same six stages without contracts and compare outcomes. The contracts are written by the system designer, so the governance floor is only as good as the contracts that define it, and the system does not yet demonstrate multi-operator governance.

Held-out stress checks. An out-of-domain logistics specimen served as a held-out stress check. In both iterations all three conditions rejected the thesis, which shows the evaluator held up outside the historical domains; it leaves precedent-memory transfer advantage unisolated, because the deterministic gates found conventional structural-kill paths first. Auxiliary historical cases (`central_station_hypothetical_target_laundering`, `central_station_mirrored_monte_carlo`) flipped across runs and serve only as variance observations.

11. Conclusion

Hardening an adversarial evaluator and governing its recursive improvement are two problems, and this paper treats them together. Failure-driven hardening, through deterministic gates and mined adversarial precedents, improves default evaluator utility on a mixed-family benchmark, while ordering effects remain exploit-family-specific and the benchmark can surface the evaluator’s own blind spots. Recursive improvement of that evaluator requires a governance layer the improving agent does not control, since an evaluator left to judge its own improvement drifts toward leniency under sustained optimization. Our meta-runner is a deliberately simple response: a deterministic orchestrator with typed promotion contracts and no learned parameters. Across six hardening stages the contracts blocked a premature promotion, scoped each stage to its named surface, and forced integration debts into separately governed programs. The architecture is deliberately narrow: it governs recursive evaluator improvement without a new optimization surface, only where the evaluation layer is itself under recursive change.

12. Future work

The evaluator pipeline emits structured adversarial traces: persuasive but structurally flawed trajectories that fail under execution, and surviving trajectories that withstand adversarial pressure. Those paired traces have the right shape for later process-supervision work in domains where objective reward signals are weak, and could in principle support supervised contrastive datasets or evaluator-training corpora. The current corpus is not yet sufficient for competitive reinforcement learning or frontier process-reward-model training. This scale only demonstrates the architectural possibility of execution-backed trace generation, and large-scale distillation of these traces into training-time systems is future work.

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Appendix A. Main-suite replication table

Only frozen full 10-specimen main-suite runs are included.

Run ID	B FA	C FA	B FR	C FR	B Good	C Good	B Struct. det.	C Struct. det.
20260405_- 090223	0.125	0.000	0.500	0.000	50.0	100.0	1.000	1.000
20260405_- 091143	0.125	0.125	0.500	0.000	50.0	100.0	0.875	0.875
20260405_- 092112	0.125	0.125	0.000	0.000	94.0	100.0	1.000	1.000
Average	0.125	0.083	0.333	0.000	64.67	100.0	0.958	0.958

Earlier N=9 runs are retained as design history and pre-freeze ablation context, but are excluded here because the frozen main benchmark now includes `t6_ai_inference_internal_price_floor` and is therefore a distinct evaluation regime.